Date: 25/07/2025

Lab Practical #8:

Study & Survey of Institute organization network infrastructure.

Practical Assignment #8:

1. Identify type of network in your institute. Draw a design of network in your institute (Any Lab/Floor/Building).

- Identify the Type of Network in Your Institute:
  - Type: LAN (Local Area Network)
  - Description: The network at Darshan University is primarily a LAN, covering individual buildings like classrooms, labs, and administrative offices within the campus. This network allows for internal communication, resource sharing, and internet access throughout the university.
- Draw a Design of the Network:
  - Selected Area: Computer Science Department Lab on the 2nd Floor
  - Network Diagram: The network in the lab is structured around a central switch that connects to all computers and network devices within the lab. The switch is connected to a router that interfaces with the university's main network and the internet. An access point provides wireless connectivity to mobile devices.
  - Diagram Elements:
    - Central Switch: Located in the network closet, connecting all lab devices.
    - Router: Connects the lab's network to the broader university network.
    - Access Point: Provides Wi-Fi coverage for students and faculty.
    - Endpoints: Desktops, printers, and other devices connected via Ethernet.

2. List how many network devices and types of cable used and give its details.

- **Network Devices:**
  - **Router:** Provides routing between the lab's network and the university's main network.
  - **Switch:** A 24-port Gigabit switch that manages connections within the lab.
  - **Access Point:** Offers wireless coverage within the lab.
- **Cables:**
  - **Cat6 Ethernet Cable:** Used for most wired connections within the lab, supports data transfer rates up to 10 Gbps, with a length limitation of 100 meters.
  - **Fiber Optic Cable:** Connects the router to the university's main network, allowing high-speed data transfer over longer distances, which is crucial for maintaining performance across the campus.